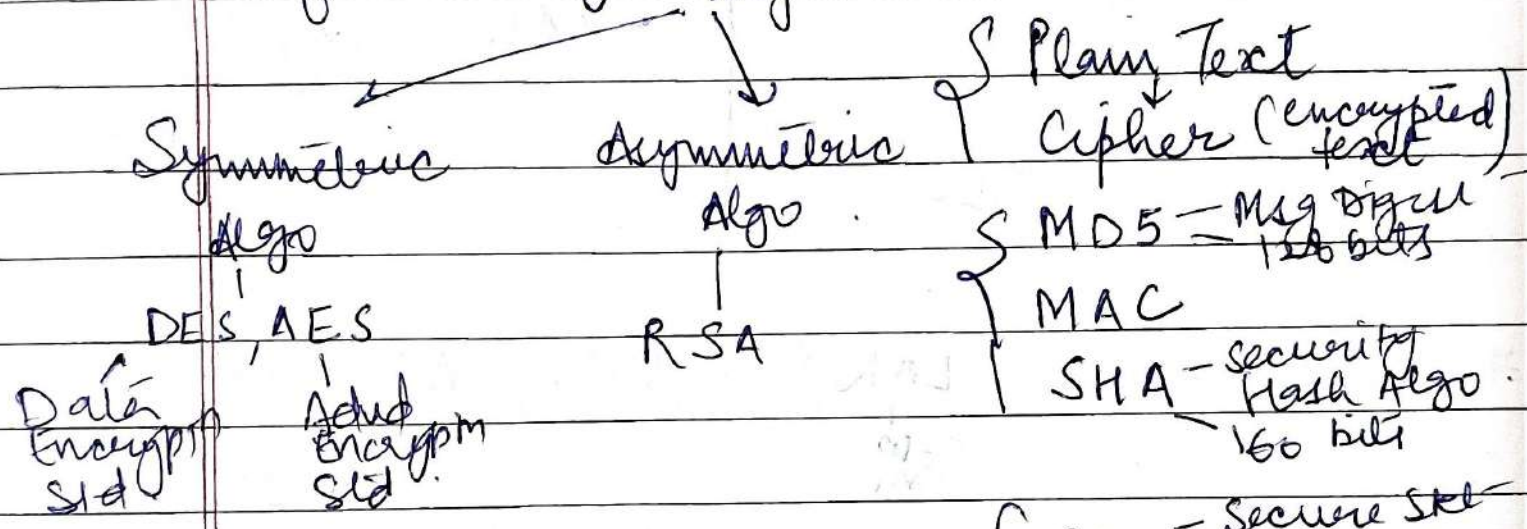


ASNS

#	Security Principles /	Books
}	Authentication - digital signature	INS - William Stallings
	Integrity - MD5, MAC	(2) Furman
	Availability	(3) Atul Kshate
	Confidentiality - cryptographic Algos	



- AL - FTP
- TL - Port 21
- NL - IP - src dest adr
- DL2 - MAC adr
- PL - bits

SSL - Secure Ssl Layer
 Secure comm b/w browser & server
 CIA

VPN
 IP Sack
 Firewall

ASNS Digital Signature

⇒ DES, RSA, DS
Confidentiality.
→ Public Key
→ Private Key.

⇒ Modular Arithmetic - add/mult inverse

eg.) $(3-5) \% 10 = (-2) \% 10 = 8$

eg) $\mathbb{Z}_{10}$
multiplicative
(2) inverse
pairs

$1 \times 1 = 1 \cdot 10 = 1$
 $5 \times 5 = 25 \cdot 10 = 5$
 $6 \times 6 = 36 \cdot 10 = 6$
 $3 \times 3 = 9 \cdot 10 = 9$
 $3 \times 7 = 21 \cdot 10 = 1$
 $9 \times 9 = 81 \cdot 10 = 1$

$11, 21, 31, 41, 51, 61, 71$
 81

eg.) $\mathbb{Z}_{26}$

(2) 3×9	(5) 11×19
(3) 5×21	(8) 17×23
(4) 7×15	

26×1 (27), 26×2 53, 26×3 79, 26×4 (105)
 $131, 156$ $3 \times 5 \times 7$

⇒ if $\gcd(a, n) = 1$, then 'a' has a multiplicative inverse in $\mathbb{Z}_n$.

$\gcd(3, 10) = \gcd(10, 7) = \gcd(7, 3) = \gcd(3, 1) = 1$.

→ $\gcd(n, 0) = n$

→ $\gcd(a, b) = \gcd(b, a)$

Steganography

ASNS

Confidentiality → Snooping, Traffic Analysis → Passive attack

Integrity (Active Atk) → Modification, Masquerading, Replaying, Repudiation, impersonation

Availability (Active Atk) → DoS

Kirchoff's Principle
in security

WMC

08.01.25

Books

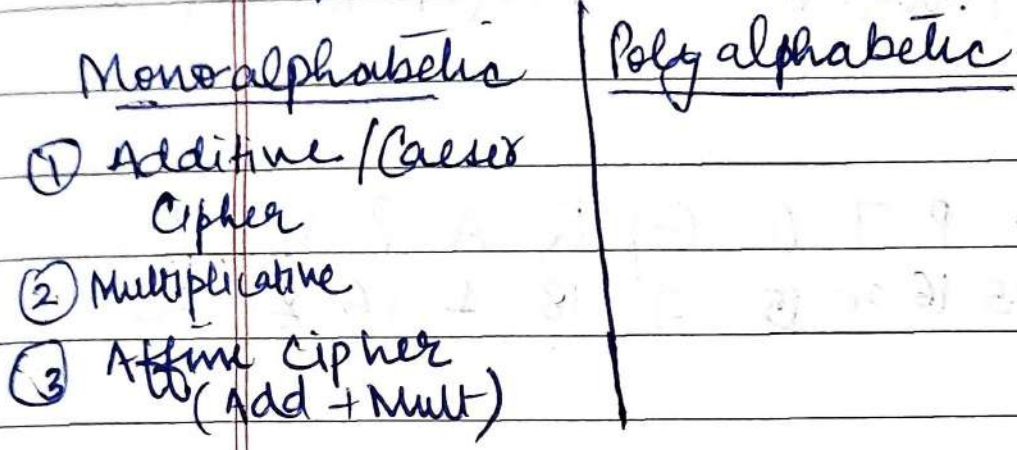
→ Mobile Commⁿ - Jochen Schille

→ Mobile Computing - R R yugal Talwar

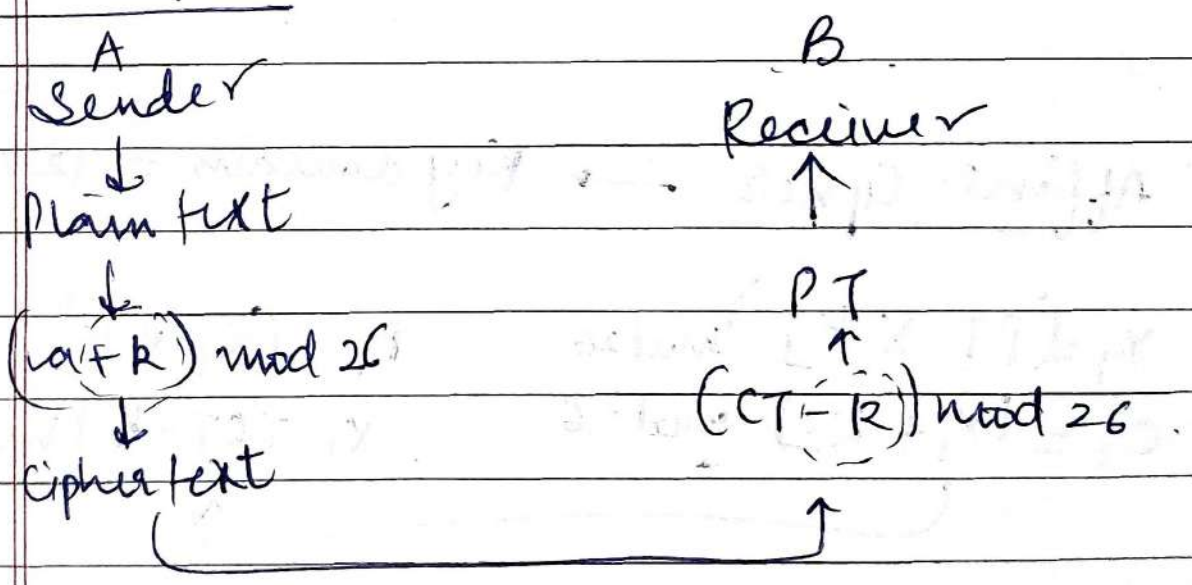
→ Silver System - Raj Pande

ASNS

⇒ Traditional ciphers Block / Stream



⇒ Caesar Cipher → Key domain = 26



eg) P = CRYPTOGRAPHY Key = 7
 C → CT = JYFWAVNYHWO
 D → PT = CRYPTOGRAPHY

⇒ Multiplicative → Key domain = 12

$$CT = (PT \times K) \bmod 26$$

$$PT = (CT \times K^{-1}) \bmod 26$$

eg.) PT = C R Y P T O G R A P H Y
 3 18 25 16 20 15 7 18 1 16 8 25
 (K=3) CT =

⇒ Affine Cipher → Key domain = 12×26

$$X_1 = (PT \times K_1) \bmod 26$$

$$CT = (X_1 + K_2) \bmod 26$$

$$PT = (X_1 \times K_1^{-1}) \bmod 26$$

$$X_1 = (CT - K_2) \bmod 26$$



ASNS

⇒

Ciphers

Substitution

Transposition

Cipher

Cipher

(Monoalphabetic cipher)

→ change position of characters

Key domain
= 26!

⇒ Polyalphabetic Cipher

① Autokey cipher

→ seq of sub keys

eg)

3 18 25 16 20 15 7 18 1 16 8 25
C R Y P T O G R A P H Y

first subkey = H₈

3 18 25 16 20 15 7 18 1 16 8

6 3 18 25 16 20 15 7 18 1 16

11 21 17 15 10 9 22 25 19 17 24

K U Q O J I V Y S Q X

A 1 P 16

B 2 Q 17

C 3 R 18

D 4 S 19

E 5 T 20

F 6 U 21

G 7 V 22

H 8 W 23

I 9 X 24

J 10 Y 25

K 11 Z 26

L 12

M 13

N 14

O 15

Encryption :

C R Y P T O G R A P H Y
 2 17 24 15 19 14 6 17 0 15 7 24

A 0 N 13 7 2 17 24 15 19 14 6 17 0 15 7
 B 1 O 14 9 19 15 13 8

C 2 P 15

D 3 Q 16

E 4 R 17

F 5 S 18

G 6 T 19

H 7 U 20

I 8 V 21

J 9 W 22

K 10 X 23

L 11 Y 24

M 12 Z 25

Decryption :

(2) Playfair cipher

5x5 matrix $\rightarrow$ A to z, I & J in single cell

$\rightarrow$ Key domain = 25!

Encryption:

PT = (C, R) (Y, P), (T, O), (G, R), (A, P), (H, Y)

K =

<u>T</u>	I/J	K	G	F
W	A	<u>H</u>	X	E
<u>Y</u>	L	<u>D</u>	<u>C</u>	Z
<u>M</u>	<u>O</u>	B	<u>R</u>	U
N	<u>P</u>	Q	<u>V</u>	S

CT = R V L N I M X V L I W D

Decryption:

PT = C R Y P T O G R A P H Y

③ Vignere Cipher
→ m-additive cipher

~~C R Y~~

PT = Y E R T E R D A Y S W E W E R E O N B U N K
Subkey = C L A S S

④ Hillfig.

ASNS

⇒ Polyalphabetic Ciphers
one to many replacement

④ Hill Cipher

- Block cipher - size $m \times m$ - key size
- matrix modular arithmetic
- residue matrix - result of modular arithmetic
- 'l' no. of blocks
- $l \times m$ - plain text chars / cipher
- multiplication process - modular
- invertible key - so can be used for decryption
 - (multiplicative inverse)
 - find det
 - find inverse
 - if in $\mathbb{Z}_n$

$$\begin{matrix} (l \times m) & \times & (m \times m) & \rightarrow & (l \times m) \\ P_T & & K & & C_T \end{matrix}$$

key domain = 26

TRANSPOSITION CIPHERS

- Keyed Ciphers — uses keys.
- Keyless Ciphers — without using keys.

(i) Railfence

eg.) PT = INFORMATION

→ I F R A I N
→ N O M T O

Keyless

(ii) Columnar Technique

eg.)
write
row
by
row

I	N	F	O
R	M	A	T
I	O	N	S

No. of cols
fixed to 4.

→ Bogus char
→ Send column
by column
→ OTSRX
→ IRIEI
→ FANUY
→ IRIEINMOCTFANUYOTSRX

(i) Permutation

→ Fix no. of columns

Key → $\begin{matrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{matrix}$ → exchange columns

→ Bit oriented cipher

→ Substitution vs Transposition

↳ same count of 0 & 1
↳ easier to decrypt

bit oriented
ciphers

Modern
ciphers

ASNS

⇒ Product ciphers - combination of invertible and non-invertible components
↳ RSA

→ Diffusion - Rⁿship b/w PT & CT

→ Confusion - Rⁿship b/w CT & K

→ Crypt analysis

Product ciphers

Fiestal

→ makes use of both invertible & non-invertible
→ eg.) DES

Non-Fiestal

→ only invertible component
→ eg.) AES

⇒ Components (PT-bits, cipher-BLK)

① P-Boxes (Permutation)

i/p lines
o/p lines

- invertible
- (i) Straight P-Box 5×5 $n \times n$
- (ii) Compression P-Box 5×3 $m \times n$
($m > n$)
- (iii) Expansion P-Box 5×7 $m \times n$
($m < n$)

(2) S-boxes

(i) 3×2 Non invertible

	00	01	10	11
00	01	11	00	10
1	00	00	01	11

Ip = 101 010 010 010

Ct = 1000 0000

(ii) 3×3 Invertible

Encryption Table

	00	01	10	11
0	111	110	101	100
1	000	010	001	011

Decryption Table

	00	01	10	11
0	100	110	101	111
1	011	010	001	000

PT = 101 010 010 010

ECCT = 010 101 101 101

dcPT = 101 010 010 010

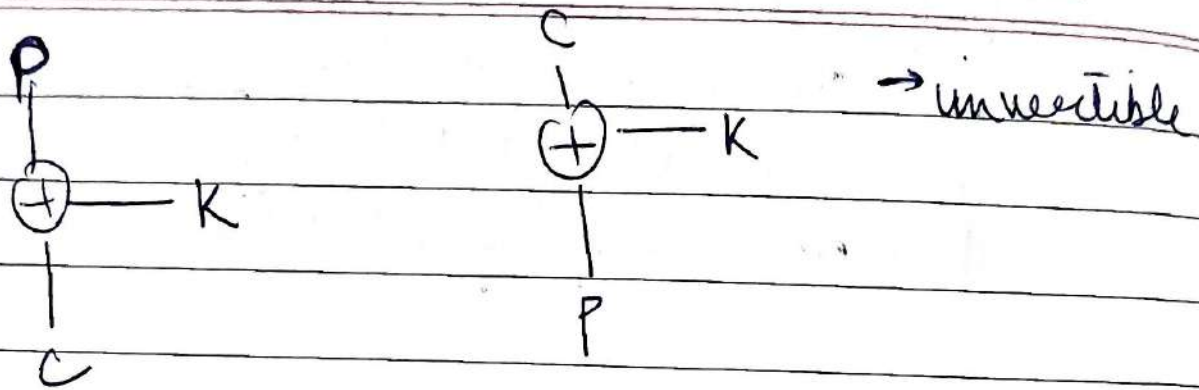
③ XOR operation

→ closure prop

→ associative

→ commutative

→ identity property - 0



- (4) Complement $\rightarrow$ invertible
- (5) Circular shift $\rightarrow$ invertible
- (6) Split and combine
- (7) Swap — when even block size
* circular shift when blk size even (n)
and key $n/2$.

ASNS

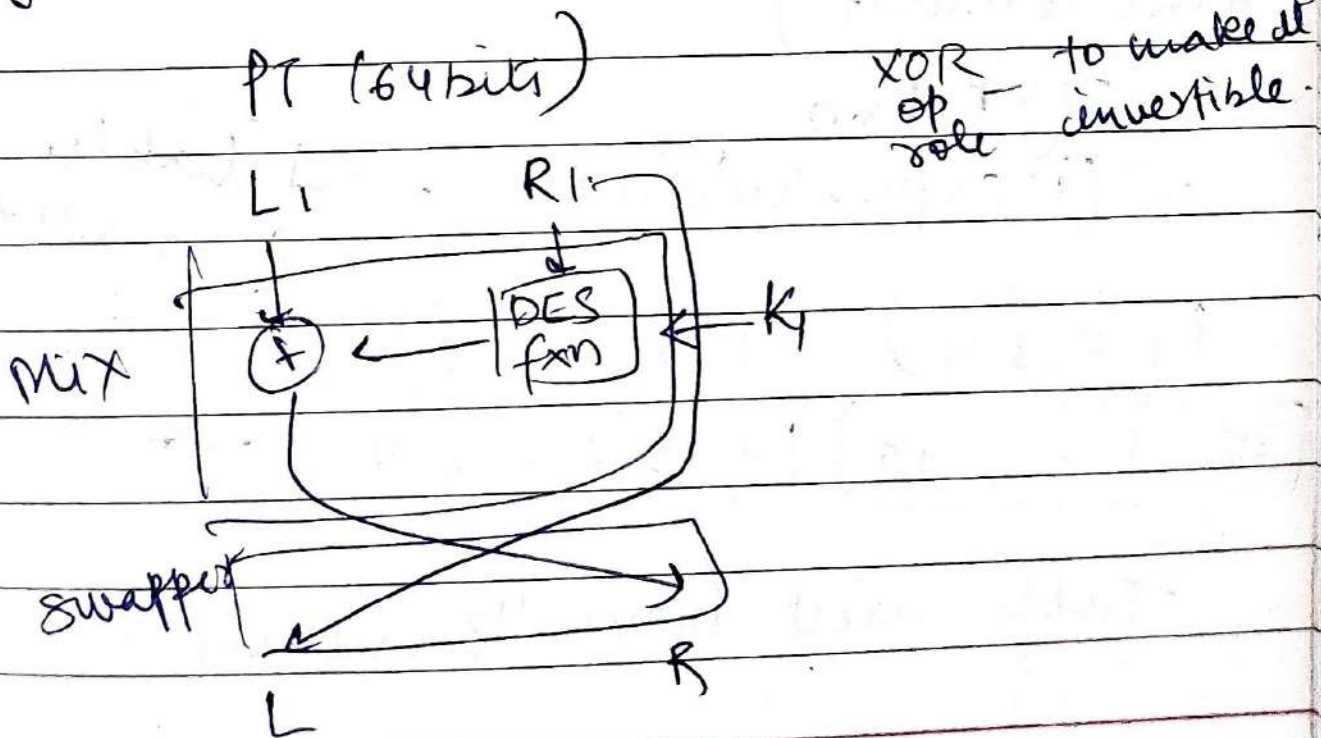
(Q) No. of padding bits $\rightarrow 0$
800 mod 64
char

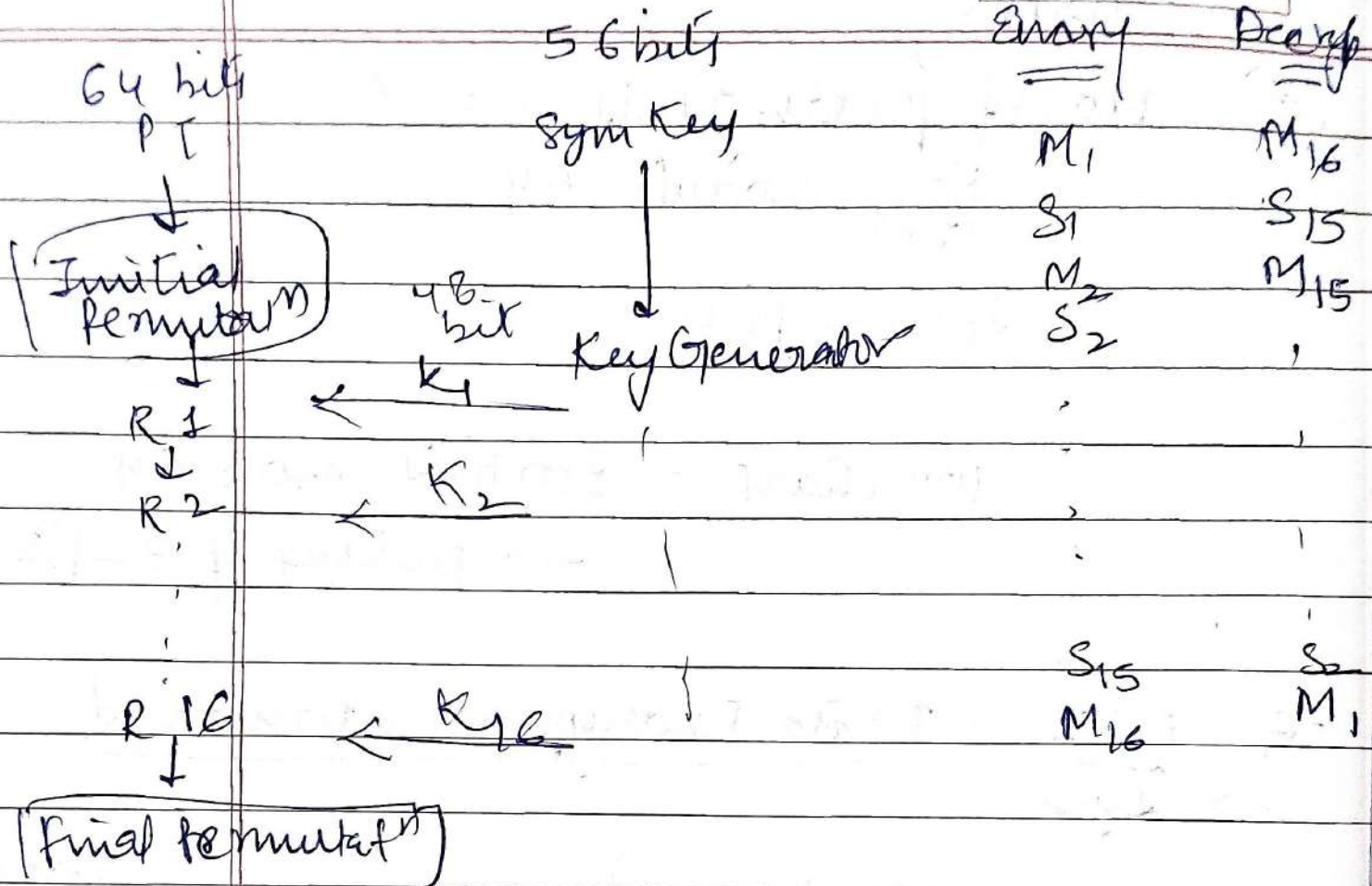
800 x 8 bits

100 char - 800 bits mod 64

$\rightarrow$ padding of 32 bits

DES - Data Encryption Standard
 $\rightarrow$ Sym





⇒ DES fxn

(i) expansion box

→ 1 table (32 × 48)

1 2 3 4	5 6 7 8	9 10 11 12
32 1 2 3 4 5	4 5 6 7 8 9	8 9 10 11 12 13

Table will have 48 entries / 48 lines

(ii) XOR / Whitener

48 bits $\oplus$ 48 bits $\rightarrow$ 48 bits
Expansion Round Key
P box
S box

8 tables (16x4)

(iii) Substitution Box 48 $\rightarrow$ 32

8 boxes (6) (6) (6) $\rightarrow$ S box 1.
16x4 table
8 boxes
(4)

(iv) Straight ~~key~~ box $\rightarrow$ 1 table (32x32)
 $\rightarrow$ 32 bits

(v)

1 Round - 2 XOR fn

6 Rounds - 32 XOR fns.

ignore every 8th bit
64 bit Key $\rightarrow$ 56 bit Key.

$\Rightarrow$ Key Generator

(i) Split $- 2 \times 28$ bits each.

(ii) Circular left on both
K fixed for round.

R 1, 2, 9, 16 $- K = 1$

Remaining rounds $- K = 2$

(iii) Combine $- 56$ bits

(iv) compression P box - 1 table
 $\rightarrow 8$ bits discarded (56×48)

$\Rightarrow$ 3-DES.

Round Key size

Assignment - Adv ES - Next Week
AES

ASNS

Same Key for encryptⁿ & decryptⁿ →

Diffie Hellman
key exchange
Algo

→ to exchange keys safely.
 → not meant for cryptography.

2 large prime no. Sender
 (not)
 [A]
 n, g

on insecure channel

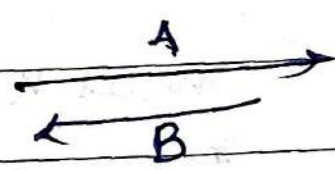
Receiver
 [B]
 n, g

Random no. generated = x — secret known to A only

y — large random no.

$$A = g^x \text{ mod } n$$

$$B = g^y \text{ mod } n$$



$$K_1 = B^x \text{ mod } n$$

$$K_2 = A^y \text{ mod } n$$

$$\underline{\underline{K_1 = K_2}}$$

eg. $n = 11, g = 7$

$x = 3$

$n = 11, g = 7$

$y = 6$

$A = g^x \text{ mod } n$

$7^3 \text{ mod } 11$

343

$A = 13 \pmod{11} = 2$

$B = g^y \text{ mod } n$

$7^6 \text{ mod } 11$

117649

7649

1049

59

$B = 4$

$K_1 = 4^3 \text{ mod } 11$

64

$K_1 = 9$

$K_2 = 2^6 \text{ mod } 11$

64

$K_2 = 9$

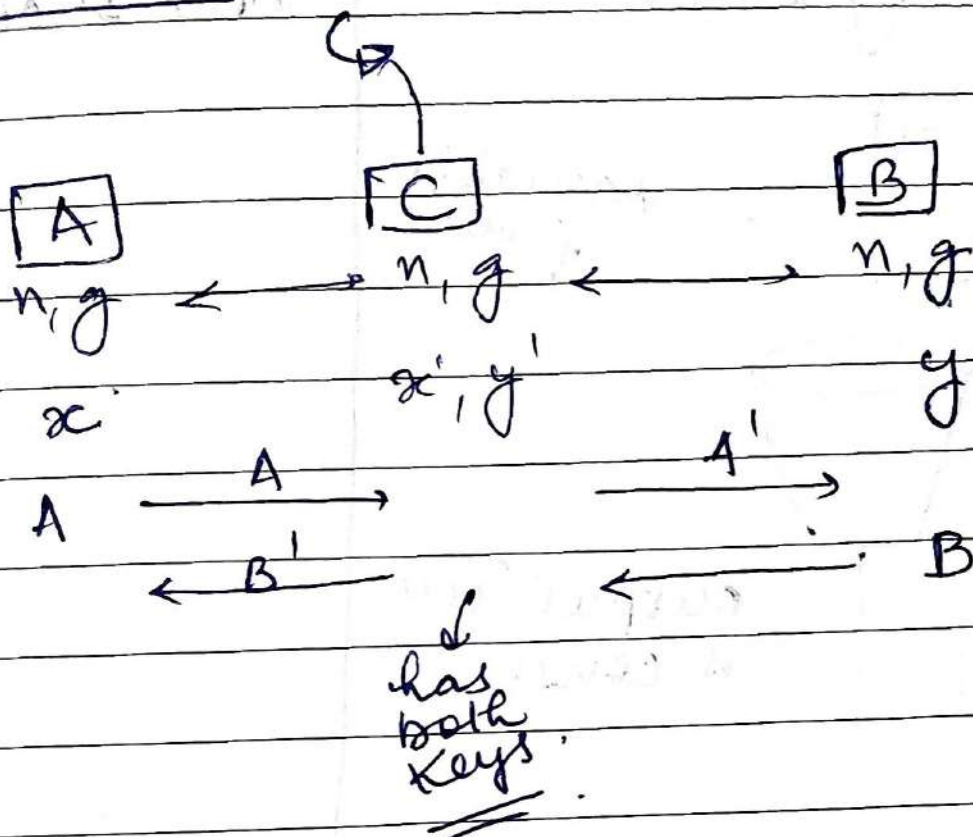
$K_1 = B^x \text{ mod } n$

$= g^{xy} \text{ mod } n$

$K_2 = A^y \text{ mod } n$

$= g^{xy} \text{ mod } n$

Drawback - Man in the Middle Attack



gives rise to asymmetric algorithm

$$\begin{array}{ccc} 11, 7 & 11, 7 & 11, 7 \\ 3 & 2, 5 & 6 \\ A = 2 & A = 7^2 \bmod 11 = 5 & B = 4 \\ B = 10 & B = 7^5 \bmod 11 = 10 & \end{array}$$

$$\begin{array}{ccc} A = 2 \xrightarrow{2} & & \xrightarrow{5} A = 5 \\ B = 10 \xleftarrow{10} & & \xleftarrow{4} B = 4 \\ K_1 = 10^3 \bmod 11 = \textcircled{1} & K_2 = 4^2 \bmod 11 = \textcircled{5} & K_2 = 5^6 \bmod 11 = \textcircled{5} \\ K_1 = 2^5 \bmod 11 = \textcircled{1} & & \end{array}$$

ASNS

RSA

→ factorizⁿ of large no. & modular
arithmetic for encrypting &
decrypting data

→ 3 main stages

- (1) key genⁿ — public & private key
- (2) encryptⁿ
- (3) decryptⁿ

→ (1) 2 large prim no p & q :-
→ kept secret

(2) Calculate the product of
prims

$$n = p \times q$$

(3.) Euler's quotient f^m

$$\phi(n) = np - q - 1$$
$$(p-1)(q-1)$$

(4) Choose encryptⁿ
v.t.

$$\rightarrow 1 < e < \phi(n)$$

$$\gcd(e, \phi(n)) = 1$$

↪ coprime

(5) Calculate decryptTM component s.t.
 $d \times e = 1 \pmod{\phi(n)}$

Cipher Text, $c = m^e \pmod{n}$

Message, $m = c^d \pmod{n}$

$$d = e^{-1} \pmod{\phi(n)}$$

ASNS

RSA

Problem with sym cryptography algo
 → how to exchange keys
 → same key used by receiver & sender
 → diffie algo
 → man in the middle problem.

→ Asym. Crypto Algo

* n —————→ $2^{2 \times n}$
 Total nodes Total Keys used

⇒ Key generation
 At node

P, q — 2 large primes

$$N = P \times Q$$

→ N will be send to the sender.

$$\phi = (P-1)(Q-1)$$

→ if intercepted diff to find P and Q from N.

→ find factors

find e s.t. factors of e $\neq$ factors of ϕ
e & ϕ are coprime

$$(exd) \bmod (p-1)(q-1) = 1$$

eg)

$p = 7$, $q = 17$
find e and d.

$$\rightarrow N = P \times Q = 7 \times 17 = 119$$

~~factors of 119 = 1, 7, 17, 119~~

$$\rightarrow \phi = (P-1)(Q-1) = 6 \times 16 = 96$$

factors of 96 = $\overset{2^0}{1}, \overset{2^1}{2}, \overset{2^2}{3}, \overset{2^3}{4}, \overset{2^4}{6}, \overset{2^5}{8}, \overset{2^6}{12}, \overset{2^7}{16}, \overset{2^8}{24}, \overset{2^9}{32}$

$$\begin{array}{r} 2 \overline{) 96} \\ 2 \overline{) 48} \\ 2 \overline{) 24} \\ 2 \overline{) 12} \\ 2 \overline{) 6} \\ 3 \overline{) 3} \\ 11 \end{array}$$

3×2^5

$$\rightarrow \boxed{e = 5}$$

$$\rightarrow (e \times d) \bmod (p-1)(q-1) = 1$$

$$(5 \times d) \bmod 96 = 1$$

$$\rightarrow 1, 97,$$

$$(96 \times k) + 1 = 5 \times d$$

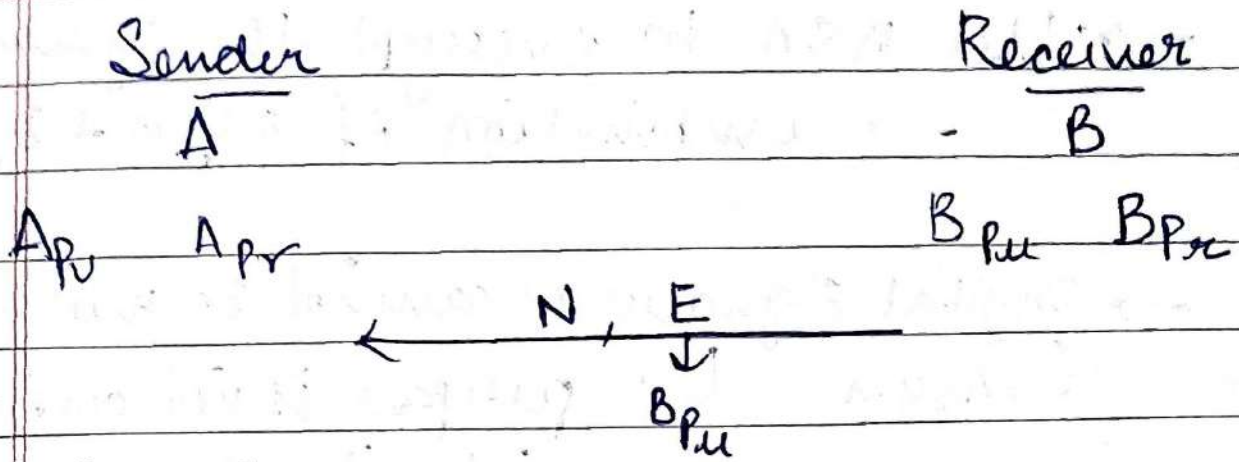
$$(96 \times 4) + 1$$

$$384 + 1$$

$$385 = 5 \times d$$

$$d = \frac{385}{5}$$

$$\boxed{d = 77}$$



Plain Text

$$CT = PT^E \text{ mod } N \xrightarrow{CT} PT = CT^D \text{ mod } N$$

eg.) $PT = ABC = 97 \ 98 \ 99$

$$CT = (97)^5 \text{ mod } 119 = 20$$

$$PT = (20)^{77} \text{ mod } 119 =$$

	Asym (RSA)	Sym (DSA)
①	Public Private Key	
②	$RSA = 2 \cdot n$	$DSA = \frac{n(n-1)}{2}$ Keys
③	speed	fast encryption & decryption
④	output bits drawback	Almost same

greater than 11p bits

→ Use RSA to encrypt the symm key
→ combination of asym + symm.

→ Digital signatures cannot be modified.
↳ Asym ↳ purpose is not confidentiality,
integrity etc.

LRC
VRC
CRC

ASNS



Message Digest

→ integrity



MD5

→ o/p is 128b — no matter the size of i/p.

$S \rightarrow \begin{bmatrix} PT \\ + MD \end{bmatrix} \rightarrow R$

(1) generator has its own MD5.

(2) Checks

$$(MD5)_R == (MD5)_S$$

if matches, integrity ✓.

*

Be able to detect if the PT has been modified!

* if $\text{size}(PT) > 2^{64}$, represent the $\text{size}(PT)$ using mod

Page No.

Date:

Steps:-

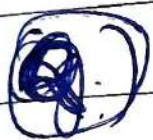
(1) Calculate the no. of bits in PT.

check 64

$$\text{size}(PT) = n(512) - 64$$

to find the size of the message.

if not, add padding bits.



$$\text{size}(PT) = 1000 \text{ b}$$

How many padding bits need to add?

$$\text{size}(PT + PB) = n(512) - 64$$

$$1000 + PB = 3 \cdot 512 - 64$$

$$PB = 472$$

→ Padding Bits = 1000....

(2) (Q)

$$1472 + PB = n \cdot 512 - 64$$

$$PB = 512 \text{ b}$$

* ipl is multiple of 512b.

↳ Blocks

Page No.

Date:

* Have to be mandatorily add:
padding bits.

(2) Calculate the size of ipl without
considering the PBs.

↳ send as 64 bits.

⇒ Consts used

(1) ABCD Reg → each 32b

↳ total 128b

gives the
128b MD5
↑ ipl

A 01 23 45 67

B 89 AB CD EF

C FE DC BA 98

D 76 54 32 10

→ initialized
with
fixed
vals.

(2) Array of ~~32 bits~~ — 64 elements
Constants / (each 32 bits)
 → represented as $t_1 - \dots - t_{64}$

→ Have 4 rounds — identical except the process used.

→ uses logical ops

only operates on BCD values
 Process P

Q

R

S

→ 512 — 16 blocks of 32 b each

Block → Subblock

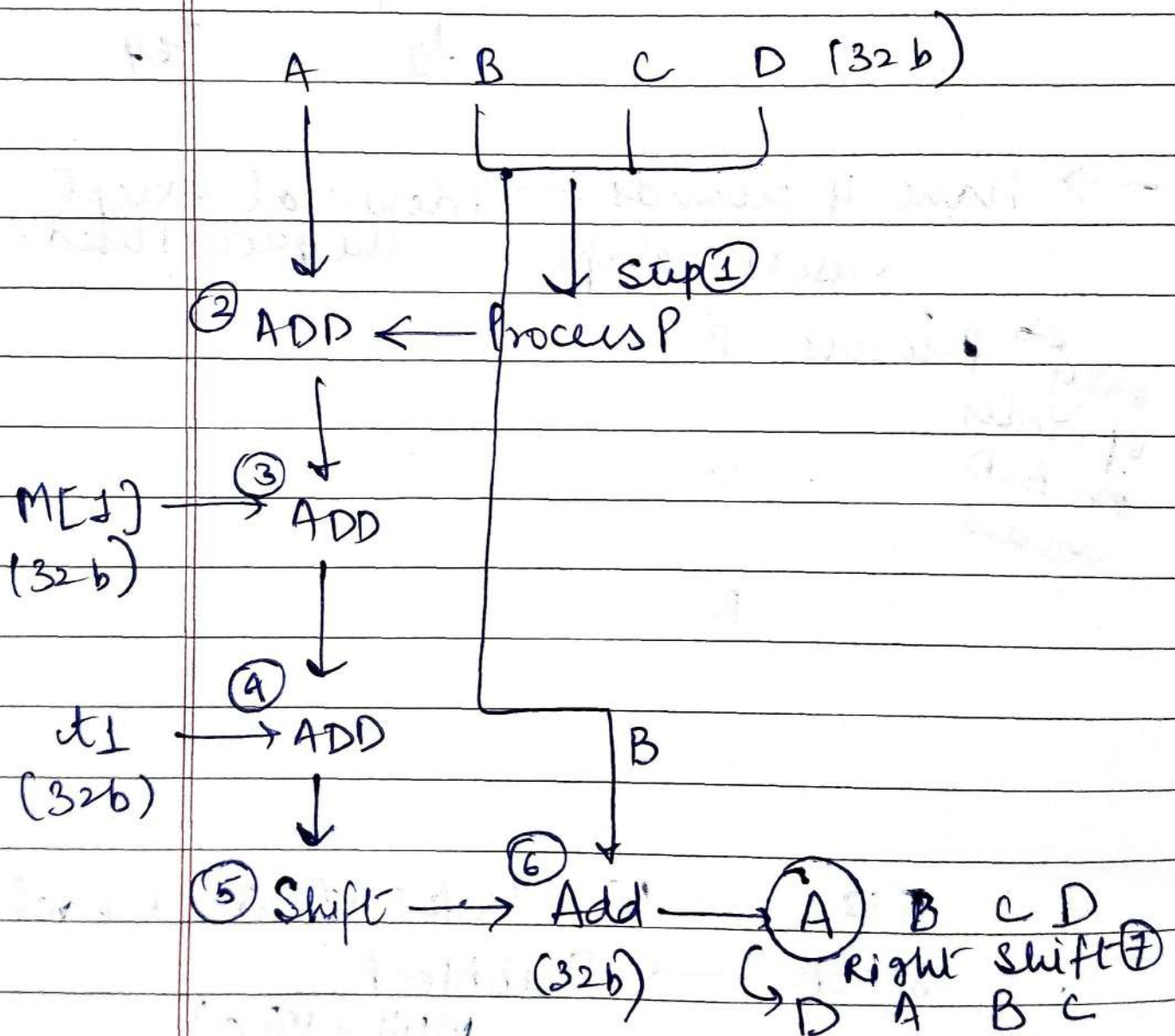
$M[1] - M[16]$

Q RSA vs Digital Signature

Q DES have an edge over RSA?

Iteration 1

$t_1, ME1$



for 1 block of 512 b.

* One of MD5 version,

Round 1 [run one process 16 times

using ME17 to ME16, d1 to d16.
→ 16 iterations

R2, R3, R4 — using diff processes
→ total 64 iterations.

⇒ Property : —

① * One directional

→ Cannot generate pf from MD.
→ non invertible.

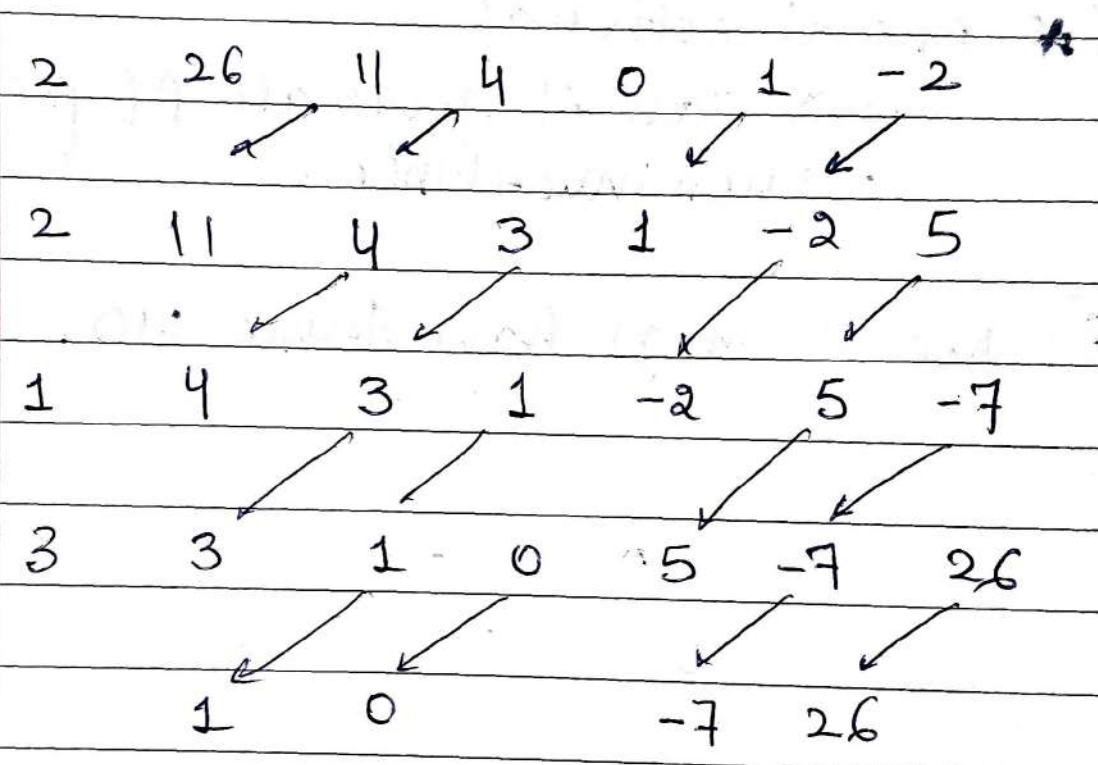
② No 2 mgs have same MD.

ASNS

⇒ Extended Euclidean

11 in $\mathbb{Z}_{26}$
 quotient remainder

$q_1, r_1, r_2, r, t_1, t_2, t$



-11 - multiplicative inverse

$$-11 + 26 = 15$$

$$x_1 = 0$$

$$x_2 = 1$$

$$x = x_1 - 2 * x_2$$

$$= 0 - 2 * 1 = -2$$

$$x_1 = 1 - 2 * (-2) = 1 + 4 = 5$$

$$x_1 = -2 - 1 * 5 = -2 - 5 = -7$$

$$x = 5 - 7 * (-3) = 26$$

(Q) find the multiplicative inverse of 7 in 26.

$$7^{-1} \pmod{26}$$

find x s.t. $7x \equiv 1 \pmod{26}$

Step 1) check if inverse exists

$$\gcd(7, 26) = 1$$

Step 2) Extended Euclidean.

$$\textcircled{26} = \textcircled{7} \times 3 + 5$$

$$7 = 5 \times 1 + 2$$

$$5 = 2 \times 2 + 1$$

$$2 = 1 \times 2 + 0$$

$$\underline{\underline{\gcd(7, 26) = 1}}$$

$$\begin{aligned} 7 \times 11 &= 77 \pmod{26} \\ &= 25 \end{aligned}$$

Q (i) I have made a toy encryptTM sys.
public mod $m = 21$

public encry key $K = 77$

RSA

message 'a' to me

Suppose that you wish to send me
the private msg 2 what is the
encrypted msg you should send?

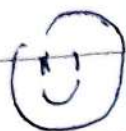
(ii) Not content with the ability to send
the put msg. you've decided to
try to read my put msg.

You find that that the dean has
send me the encrypted msg

95. What was the dean's
actual msg to me?

(i)

ASNS



MESSAGE AUTHENTICATION CODE (MAC)

Hash based
* HMAC

integrity & Authentication

- uses shared symm. key
- at both ends performs encryptⁿ
 $(MAC)_{\text{sender}} == (MAC)_{\text{receiver}}$
 ↳ to check if PT modified.

▷ Beacuz key was same,
if mac matches,
then that means S has the
key → user verified.

⇒ HMAC → ^{8 bits} 00110110 = 36 in Hexdec.
 ipad → inner padding bits
 opad → outer " "
 ↳ 01011100 = 5c

→ repeated $\frac{b}{8}$ times

where $b = \text{blk size}$

→ total ipad bits = b
if repeated $\frac{b}{8}$ times
 b bits of ipad.

→ Shared secret Key - K

Size of K - mutually agreed by S & R

PT - block size - fixed for a particular MAC algo.

~~Step 1~~ Req 1. — $b = K$

① if $b > k$ — add padding to K .
on left side

eg.) $k = 8b$

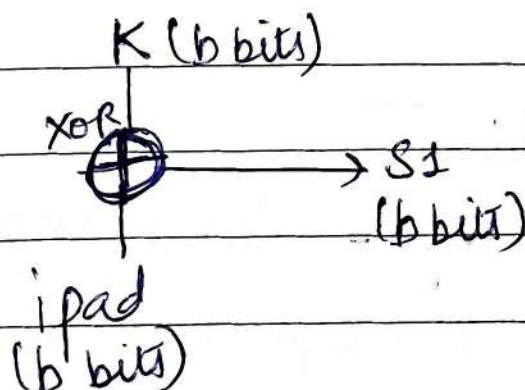
$b = 64b$

$k = \underbrace{000 \dots 0}_{56 \text{ times}} \underbrace{1}_{64 \text{ times}} \text{ Key}$
padding

(2) if $b < k \rightarrow$ truncate K .
by using suitable MD5

eg.) compression P-box — invertible.

Step 2

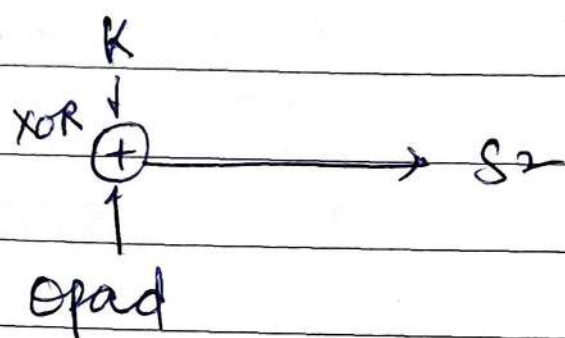


Step 3

$(S1) \oplus (M) \rightarrow \text{append}$
(b bits)

Step 4 Apply MD Alg to generate $H1$.

Step 5



Step 6

$(H1) \text{ f } S2$ (append)

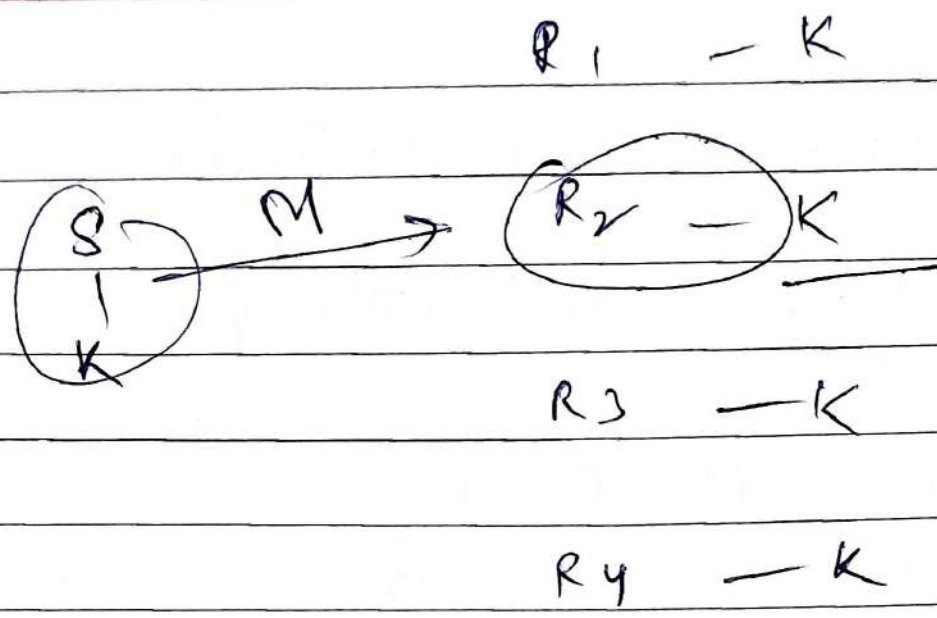
Step 7

Apply suitable MD5 algo to generate final $H2$ called HMAC.

Problems: —

- ① Key sharing — then asym key — RSA.
- ② 1 sender multiple receivers.
- ③ problem of repudiation is possible.

outside
in
ing)



How does
R2 know
that msg
has been
sent by
S not R1
or R3/R4

→ Authentication fails

ASNS

Page No.

Date: 10.2.26



IDEA - International Data

Encryption Algorithm

→ proposed in 1991

earlier called PES (Proposed Encryption Standard)

Features

(1)

Sym Key

cryptographic algo - $\text{encrypt}^n + \text{decrypt}^n$

(2)

Block cipher

→ size of blk = 64 bits

(3)

Key size = 128 bits ~~→~~ ⁸

(4)

Rounds = 8

→ All rounds are identical

8 rounds followed
by o/p transformation

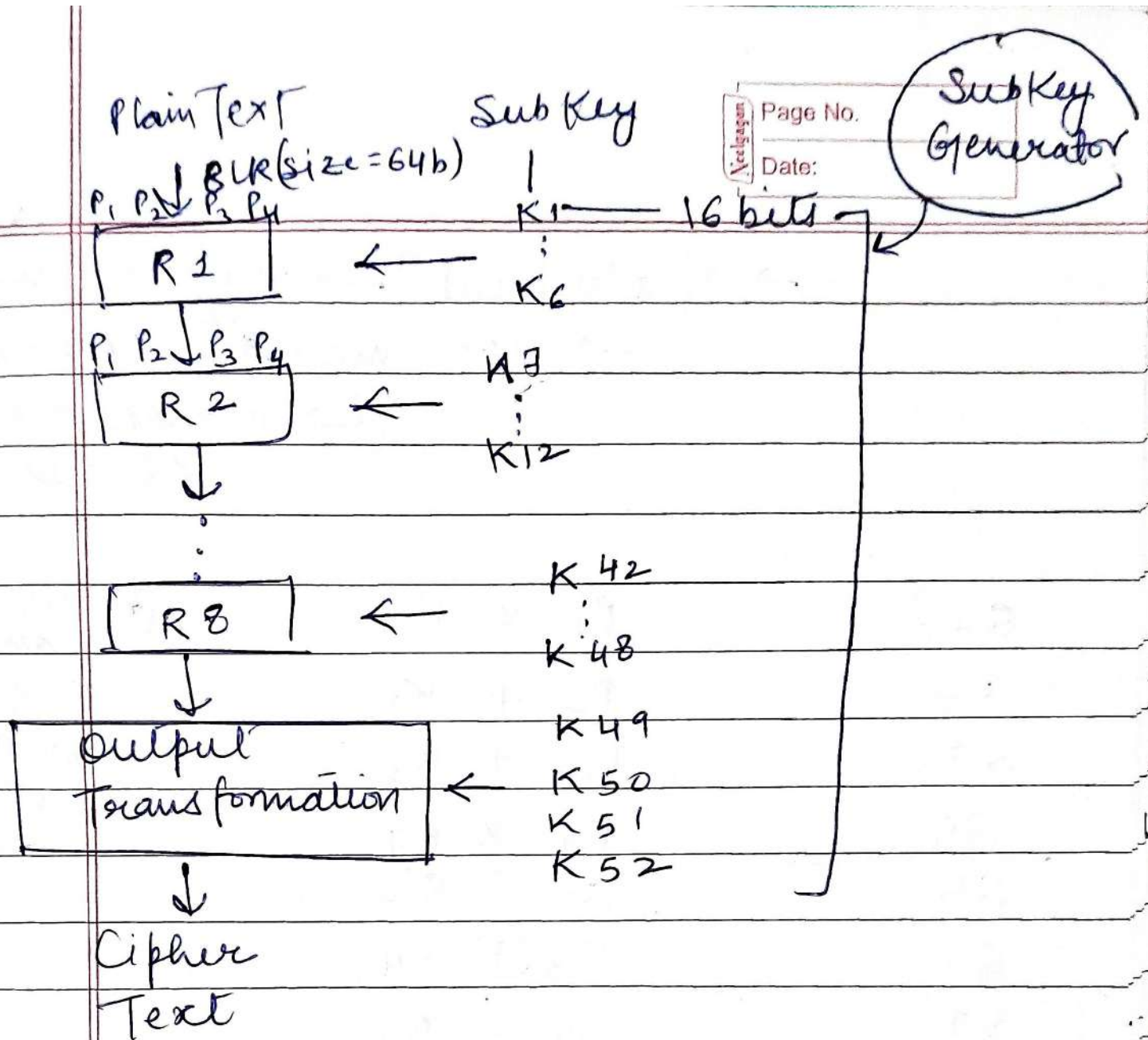
→ Each round 6 subkeys are
used

→ total 48 subkeys

+ 4 subkey for o/p
transformation

= 52 Subkeys

DES
64
64
16



Step 1) BLK 64 b $\rightarrow$ 4 parts of 16 b each
 $B_1 \rightarrow P_1, P_2, P_3, P_4$

Step 2) Round 1 $\rightarrow$ 14 steps
 $\rightarrow$ uses addⁿ, multipⁿ, XOR
 $\rightarrow$ binary opⁿs.

(> 16 bits)
 $\rightarrow$ if add / multⁿ $\rightarrow$ more bits in op
 $\rightarrow$ use mod 2^{16} (65536)
 $\hookrightarrow$ to make op 16 bits

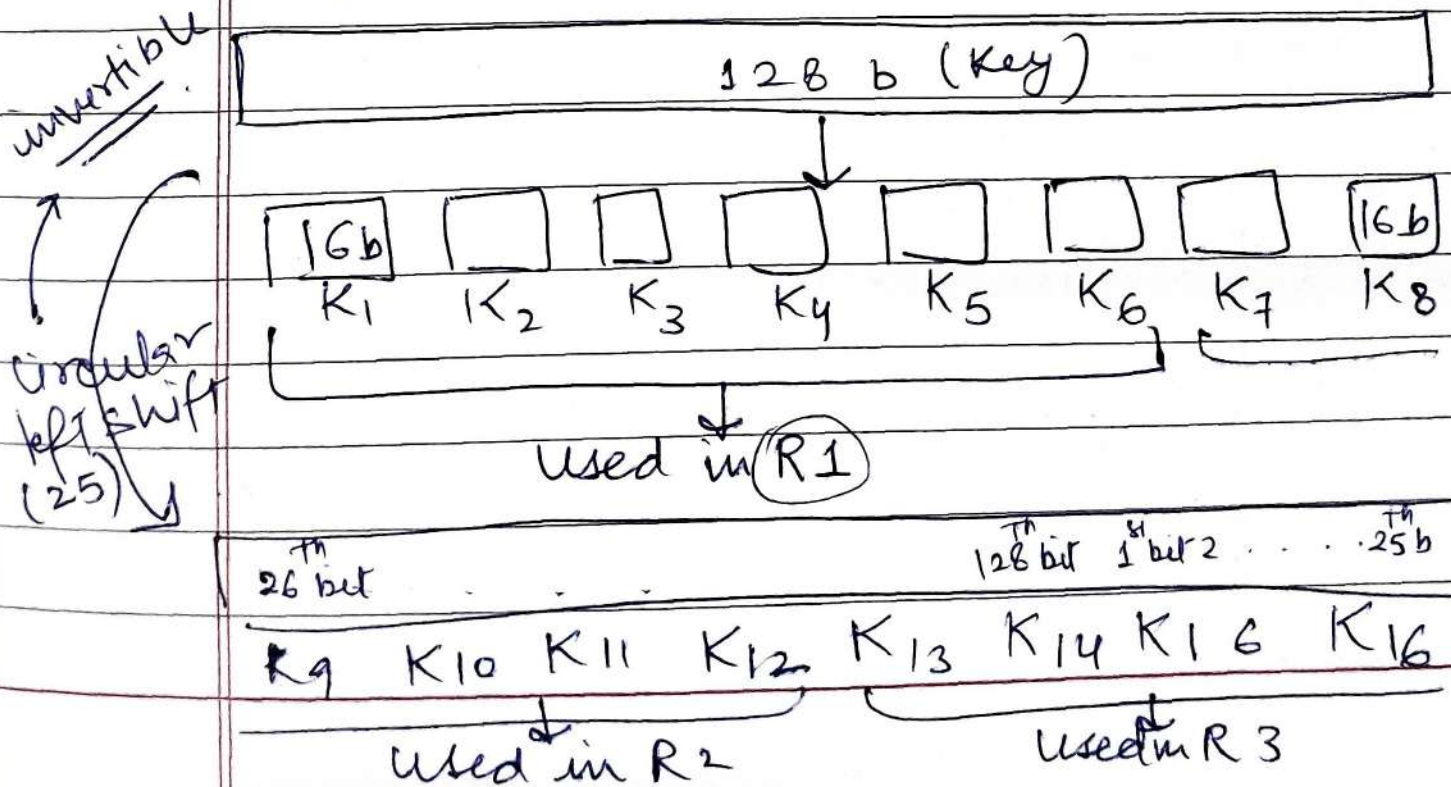
S1	$P_1 \times K_1$	(Q) How many $\oplus$ op ⁿ , add op ⁿ etc.
S2	$P_2 + K_2$	
S3	$P_3 + K_3$	
S4	$P_4 \times K_4$	
S5	$S_1 \oplus S_2$	
S6	$S_3 \oplus S_4$	
S7	$S_5 \times K_5$	
S8	$S_6 + S_7$	
S9	$S_8 \times K_6$	
S10	$S_7 + S_9$	
S11	$S_1 \oplus S_9 \rightarrow P_1$	
S12	$S_3 \oplus S_9 \rightarrow P_2$	Swap
S13	$S_2 \oplus S_{10} \rightarrow P_3$	
S14	$S_4 \oplus S_{10} \rightarrow P_4$	

* In Round 8, do not swap P_2 & P_3

⇒ Output Transformation

$$\left. \begin{array}{l} P_1 \times K_{49} = C_1 - (16b) \\ P_2 + K_{50} = C_2 \\ P_3 + K_{51} = C_3 \\ P_4 \times K_{52} = C_4 \end{array} \right\} \begin{array}{l} 64b \\ \text{Cipher} \\ \text{Text} \end{array}$$

⇒ Key Generation



and so on...

→ diffusion
→ confusion

ASNS

Confidentiality
Integrity
Authentication
Known reputation

Page No.

Date: 11.02.26

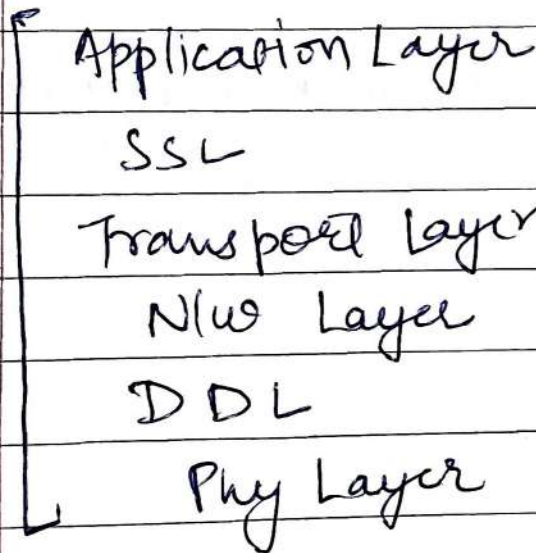


SSL (Secure Socket Layer)

→ secure commⁿ b/w web server & client

→ uses both sym & asym cryptography + MAC + MD5

TCP
model



⇒ Combination of 3 protocols: -

- (1) Handshake protocol
- (2) Record protocol
- (3) Alert protocol

→ When handshake protocol is successful, then moves to record protocol...

⇒ Handshake includes 4 steps :-

- (1) Establishing security capabilities
- (2) Server authentication and key exchange
- (3) Client authentication and key exchange
- (4) finished

[1] Establishing security capabilities :-

msg { type (1B)
 + len (2B)
 + content (NB)

① Client sends "Hello" msg to Server.
 [Client initiates]

in msg content {
 → Version of SSL — highest version of SSL that client can support.
 [32B]
 → Random No — Timestamp [4B] + Random no [28B]
 → Session ID — int value — 0 or non zero
 ↳ already created cnxn
 ↳ req for 1st cnxn or new cnxn
 → Compression suite — list of compression algo that client supports
 → Cryptographic algo — DES, 3-DES etc that client can support
 }
 Graceful termination — can start the same cnxn
 Abort termination — start new cnxn
 If time exceeded 24 hrs — start new cnxn
 ↳ uses client timestamp.

② Server replies

→ Version — highest version that client can support and client server is comfortable with
 [32B]
 → Random No — timestamp [4B] + Random no [28B]
 → Session ID — for client SID = 0, generate & pass SID
 for client SID != 0, if server OK, then return same SID
 → Compression suite — mutually agreed.
 → Cryptographic algo — mutually agreed.

* Server authentication is mandatory
→ But server can require that client is authenticated.

[2]

Server authⁿ :-

Assign Key

Server → sole sender
sends :-

- ① digital certificate — ^{has} public key
- ② if doesn't have digital certificate,
(share public key.
also sends no certificate msg.
- ③ Req client auth — ^{ask for} certificate
- ④ done msg

[3]

Client authⁿ :-

Client - sole sender

- ① if requested by server & has certificate, then send certificate.
→ else send no certificate.

② Generate 48B pre-master secret.

↳ encrypted with server's public key.

Both client & server generate

Pre master + RN of client + RN of server

↓ MD Algo

Master Secret

↳ shared secret key used in MAC

↳ integrity

(4) finished! —

① Master Secret + RN client + RN server } Both C & S
↓ MD Algo
Sym Key

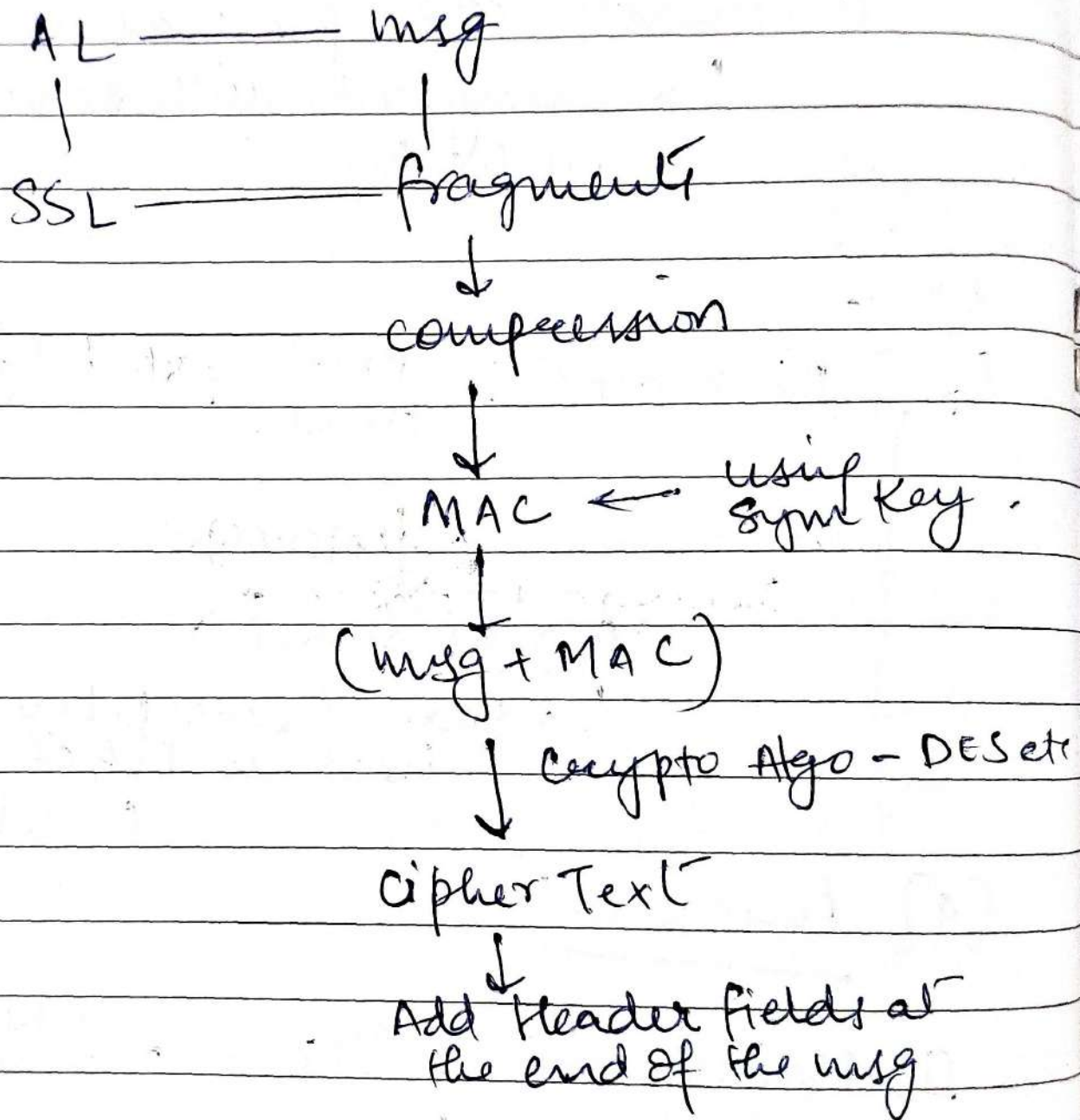
⇒ Asym used to share Sym Key

⇒ Record Protocol : —

Sender

Page No.

Date:

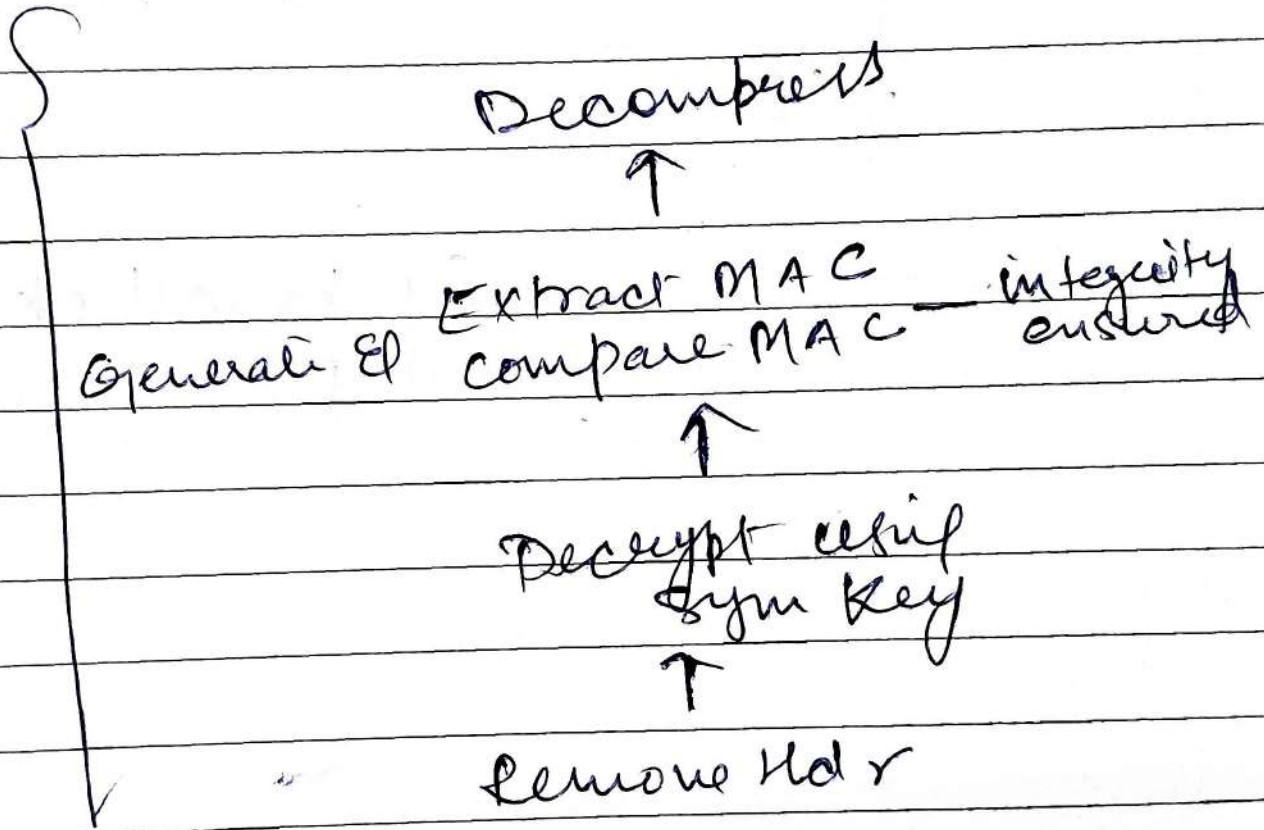


Received

Page No.

Date: ✓

SSL



⇒ Alert protocol : —

2B Alert msgs are exchanged

{ 1B - Type - 0 - critical
2 - ~~Warning~~ Warning
+ 1B

→ Either during handshake
or record protocol